SHUCHANG LIU

Email: shucliu@mit.edu

RESEARCH EXPERIENCE

Massachusetts Institute of Technology Cambridge, USA
Postdoctoral researcher Jun. 2024 – present

- A physically-constrained downscaling approach based on dynamical downscaling and machine learning **[Funded by Swiss National Science Foundation Postdoc Mobility fellowship, CHF 128k]**
- Constraining precipitation uncertainty under climate change using machine learning (MIT Climate Grand Challenges Initiative)
- Generalized convection parameterization: adapting current-climate trained models for warmer climates (Affiliate scientist of Multiscale Machine Learning In Coupled Earth System Modeling)

ETH Zürich Zürich, Switzerland
Postdoctoral researcher Oct. 2023 – May. 2024

- How to deal with double-ITCZ when downscaling in the tropics

PhD candidate Oct. 2020 – Sep. 2023

- The potential of high-resolution climate models in the tropics in constraining climate-change uncertainties

Tsinghua University Beijing, China
Student Researcher Sep. 2017 - June 2020

- The impact of climate mitigation pathways on the heatwave and air pollution-related human health in China

EDUCATION

ETH Zürich Zürich, Switzerland
PhD Atmospheric and Climate Science Sep. 2023
Tsinghua University Beijing, China
M.S. Environmental Sciences and Engineering; outstanding graduate June 2020
Nanjing University Nanjing, China
B.S. Environmental Engineering; outstanding graduate July 2017

PUBLICATIONS

H-index: 13, Citations: 753

JOURNAL

Liu, S., Paul A. O'Gorman. CERA: A Framework for Improved Generalization of Machine Learning Models to Changed Climates. *Under revision. Journal of Advances in Modeling Earth Systems.* (2025) https://arxiv.org/pdf/2509.00010

Liu, S., Zeman, C., & Schär, C. Assessing cloud feedbacks over the Atlantic with bias-corrected downscaling. *Journal of Advances in Modeling Earth Systems (impact factor: 7.8)*, 17, (2025): e2024MS004661. https://doi.org/10.1029/2024MS004661

Wang, Y., Liu, Y., **Liu, S.**, Wang, B., Duan, X., Wang, S., & Zhao, B. Changing climate alters the time people spend outdoors. *Sustainable Horizons (impact factor: 5.5)*, (2025): 13, 100133. https://doi.org/10.1016/j.horiz.2025.10013

Liu, S., Zeman, C., & Schär, C. Dynamical Downscaling of Climate Simulations in the Tropics. *Geophysical Research Letters (impact factor: 5.1)*, 51, (2024): e2023GL105733. https://doi.org/10.1029/2023GL105733

Yao, M., Niu, Y., **Liu, S.**, Liu, Y., Kan, H., Wang, S., ... & Zhao, B. Mortality burden of cardiovascular disease attributable to ozone in China: 2019 vs 2050. *Environmental Science & Technology (impact factor: 12.4)*, (2023): 57(30), 10985-10997. https://doi.org/10.1021/acs.est.3c02076

Jiang, Y., Ding, D., Dong, Z., **Liu, S.**, Chang, X., Zheng, H., ... & Wang, S. Extreme emission reduction requirements for china to achieve world health organization global air quality guidelines. *Environmental Science & Technology (impact factor: 12.4)*, (2023): 57(11), 4424-4433. https://doi.org/10.1021/acs.est.2c09164

Liu, S., Zeman, C., Sørland, S. L., & Schär, C. Systematic Calibration of a Convection-Resolving Model: Application over Tropical Atlantic. *Journal of Geophysical Research: Atmospheres (impact factor: 4.4)*, (2022): e2022JD037303. https://doi.org/10.1029/2022JD037303

Liu, Y., **Liu, S.**, Wang, S., & Zhao, B. How will window opening change under global warming: A study for China residence. *Building and Environment (impact factor: 7,9)*, (2022): 209, 108672. https://doi.org/10.1016/j.buildenv.2021.108672

Liu, S., Xing, J., Wang, S., Ding, D., Cui, Y., & Hao, J. Health benefits of emission reduction under 1.5℃ pathways far outweigh climate-related variations in China. *Environmental Science & Technology (impact factor: 12.4)*, (2021): 55(16), 10957-10966. https://doi.org/10.1021/acs.est.1c01583

Liu, S., Xing, J., Sahu, S. K., Liu, X., **Liu, S.**, Jiang, Y., ... & Wang, S. Wind-blown dust and its impacts on particulate matter pollution in Northern China: current and future scenarios. *Environmental Research Letters (impact factor: 7.2)*, (2021): 16(11), 114041. https://doi.org/10.1088/1748-9326/ac31ec

Ding, D., Xing, J., Wang, S., Dong, Z., Zhang, F., **Liu, S.**, & Hao, J. Optimization of a NOx and VOC cooperative control strategy based on clean air benefits. *Environmental science & Technology (impact factor: 12.4)*, (2021): 56(2), 739-749. https://doi.org/10.1021/acs.est.1c04201

Sahu, S. K., **Liu, S.,** Liu, S., Ding, D., & Xing, J. Ozone pollution in China: Background and transboundary contributions to ozone concentration & related health effects across the country. *Science of the Total Environment (impact factor: 8.7)*, (2021): 761, 144131. https://doi.org/10.1016/j.scitotenv.2020.144131

Jiang, Y., Xing, J., Wang, S., Chang, X., **Liu, S.,** Shi, A., ... & Sahu, S. K. Understand the local and regional contributions on air pollution from the view of human health impacts. *Frontiers of Environmental Science & Engineering (impact factor: 5.5)*, (2021): 15(5), 88. https://doi.org/10.1007/s11783-020-1382-2

Liu, S., Xing, J., Westervelt, D. M., **Liu, S.,** Ding, D., Fiore, A. M., ... & Wang, S. Role of emission controls in reducing the 2050 climate change penalty for $PM_{2.5}$ in China. *Science of the Total Environment (impact factor: 8.7)*, (2021): 765, 144338. https://doi.org/10.1016/j.scitotenv.2020.144338

Liu, S., Xing, J., Wang, S., Ding, D., Chen, L., Hao, J. Revealing the impacts of transboundary pollution on $PM_{2.5}$-related deaths in China. *Environment International (impact factor: 11.6)*, (2020): 134, 105323. https://doi.org/10.1016/j.envint.2019.105323

Xing, J., Lu, X., Wang, S., Wang, T., Ding, D., Yu, S., ... & Hao, J. The quest for improved air quality may push China to continue its CO_2 reduction beyond the Paris Commitment. *Proceedings of the National Academy of Sciences (impact factor: 10.6)*, (2020): 117(47), 29535-29542. https://doi.org/10.1073/pnas.2013297117

Chen, L., Xing, J., Mathur, R., **Liu, S.,** Wang, S., & Hao, J. Quantification of the enhancement of $PM_{2.5}$ concentration by the downward transport of ozone from the stratosphere. *Chemosphere (impact factor: 7.7)*, (2020): 255, 126907. https://doi.org/10.1016/j.chemosphere.2020.126907

Sahu, S. K., Chen, L., **Liu, S.,** Ding, D., & Xing, J. The impact of aerosol direct radiative effects on $PM_{2.5}$-related health risk in Northern Hemisphere during 2013–2017. *Chemosphere (impact factor: 7.7)*, (2020): 254, 126832. https://doi.org/10.1016/j.chemosphere.2020.126832

BOOK

Wang, S., **Liu, S.** Air pollution and lung cancer risks. In: Nriagu, J.O. (Ed.), *Encyclopaedia of environmental health, 2nd Edition*. Elsevier Science. (2019). https://doi.org/10.1016/B978-0-12-409548-9.11823-8

PRESENTATIONS

Dec. 2025	American Geophysical Union Meeting (Talk), New Orleans, USA
Nov. 2025	Invited talk at Princeton University, USA
Oct. 2025	Invited talk by Machine Learning and Climate Meetups in the Greater Boston area, MIT, USA
Oct. 2025	Invited talk by Multiscale Machine Learning In Coupled Earth System Modeling, NYU, USA
July 2025	Invited talk at University of Lausanne, Switzerland
Jan. 2025	Invited talk at ETH Zürich, Switzerland
Dec. 2024	American Geophysical Union Meeting (Poster), Washington, D.C., USA
Dec. 2023	American Geophysical Union Meeting (Talk), San Francisco, USA
April 2023	European Geophysical Union Meeting (Talk), Vienna, Austria
May 2022	European Geophysical Union Meeting (Talk), Vienna, Austria

TEACHING & SUPERVISING EXPERIENCES

Bachelor's Courses

Risk analysis of environmental health — Tsinghua University

Numerical method in environmental physics, Environmental Systems, Python in Geosciences — ETH Zürich

Master's Courses

Numerical modeling of weather and climate models — ETH Zürich

Bachelor's Thesis

Visualization of simulated clouds over the Atlantic — ETH Zürich

Master's Thesis

Extreme Heat Waves Over the Persian Gulf — ETH Zürich

SERVICES

Reviewer for *Earth's Future, Journal of Advances in Modeling Earth Systems, Geophysical Research Letters, and Artificial Intelligence for the Earth Systems* — Since 2025

Organizer of monthly *Machine Learning and Climate Meetups* in the Greater Boston area. — Since 2025

(Engaging participants from MIT, Harvard, Google, Boston University, and Boston College across all career stages.)